LAPORAN PRAKTIKUM MONGODB
Laporan Pertemuan 6

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TEKNOLOGI REKAYASA INFORMATIKA INDUSTRI

POLITEKNIK MANUFAKTUR BANDUNG

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Pendahuluan

Praktikum ini bertujuan untuk memahami dasar operasi MongoDB khususnya dalam pengolahan data menggunakan aggregation pipeline. Mahasiswa diharapkan mampu melakukan pembuatan data dummy, pengolahan data produksi dan inspeksi, serta menganalisis performa mesin melalui perhitungan metrik seperti OEE (Overall Equipment Effectiveness). Selain itu, praktikum ini juga melatih kemampuan dalam mengolah data skala besar untuk menghasilkan informasi yang berguna dalam pengambilan keputusan di lingkungan industri.

Desain Data

Diagram Koleksi sensor

Field	Tipe Data	Contoh Nilai	Keterangan
mesin	String	CNC-01	ID mesin
suhu	Float	85.5	Suhu mesin (°C)
getaran	Float	0.32	Tingkat getaran
timestamp	Date	2026-05-06T08:20:00	Waktu pencatatan data
status	String	normal / maintenance	Status kondisi mesin

Diagram Koleksi maintenance

Field	Tipe Data	Contoh Nilai	Keterangan
mesin	String	CNC-01	ID mesin
tanggal	Date	2024-01-10	Tanggal maintenance
biaya	Integer	1200000	Biaya perawatan
teknisi	String	Andi	Nama teknisi

KEGIATAN PRAKTIKUM TERBIMBING

1. Membuat file koneksi.py, isi dengan kode berikut:

```
import os
from dotenv import load_dotenv
from pymongo import MongoClient

load_dotenv()
client = MongoClient(os.getenv("MONGO_URI"),serverSelectionTimeoutMS=5000)

try:
    client.admin.command('ping')
    print("Koneksi MongoDB berhasil!")
except Exception as e:
    print("Gagal terhubung:", e)
finally:
    client.close()
```

2. Membuat file crud_sensor.py lalu isi dengan kode berikut:

```
from dotenv import load_dotenv
from pymongo import MongoClient
from datetime import datetime, timedelta
import random, os

load_dotenv()
client = MongoClient(os.getenv("MONGO_URI"))
db = client[os.getenv("DB_NAME")]
collection = db["sensor"]
```

3. Membuat file insertmany.py dan isi dengan kode berikut:

```
from dotenv import load_dotenv
from pymongo import MongoClient
from datetime import datetime, timedelta
import random, os

load_dotenv()
client = MongoClient(os.getenv("MONGO_URI"))
db = client[os.getenv("DB_NAME")]
collection = db["sensor"]

data = []
for i in range(10):
    doc = {
        "mesin": f"CNC-{random.randint(1,3):02d}",
        "suhu": round(random.uniform(60, 100), 2),
```

```
        "getaran": round(random.uniform(0.1, 0.5), 2),
        "timestamp": datetime.utcnow() - timedelta(minutes=i*5),
        "status": "normal"
    }
    data.append(doc)
result = collection.insert_many(data)
print(f"Tersimpan {len(result.inserted_ids)} dokumen")
```

Output:

```
(venv) PS C:\Users\mdhaf\Downloads\praktikum6> python -u "c:\Users\mdhaf\
c:\Users\mdhaf\Downloads\praktikum6\insertmany.py:17: DeprecationWarning:
resent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  "timestamp": datetime.utcnow() - timedelta(minutes=i*5),
Tersimpan 10 dokumen
```

```
_id: ObjectId('69f9b71982818aa2b309ad1b')
mesin : "CNC-01"
suhu : 64
getaran : 0.18
timestamp : 2026-05-05T09:23:37.058+00:00
status : "normal"

_id: ObjectId('69f9b71982818aa2b309ad1c')
mesin : "CNC-02"
suhu : 96.36
getaran : 0.32
timestamp : 2026-05-05T09:18:37.058+00:00
status : "normal"

_id: ObjectId('69f9b71982818aa2b309ad1d')
mesin : "CNC-01"
suhu : 97.24
getaran : 0.26
timestamp : 2026-05-05T09:13:37.058+00:00
status : "normal"

_id: ObjectId('69f9b71982818aa2b309ad1e')
mesin : "CNC-02"
suhu : 62.93
getaran : 0.42
timestamp : 2026-05-05T09:08:37.058+00:00
status : "normal"

_id: ObjectId('69f9b71982818aa2b309ad1f')
mesin : "CNC-01"
suhu : 69.95
getaran : 0.17
timestamp : 2026-05-05T09:03:37.058+00:00
status : "normal"

_id: ObjectId('69f9b71982818aa2b309ad20')
mesin : "CNC-01"
suhu : 90.12
getaran : 0.49
timestamp : 2026-05-05T08:58:37.058+00:00
status : "normal"
```

```
_id: ObjectId('69f9b71982818aa2b309ad21')
mesin : "CNC-03"
suhu : 93.28
getaran : 0.33
timestamp : 2026-05-05T08:53:37.058+00:00
status : "normal"

_id: ObjectId('69f9b71982818aa2b309ad22')
mesin : "CNC-02"
suhu : 74.71
getaran : 0.24
timestamp : 2026-05-05T08:48:37.058+00:00
status : "normal"

_id: ObjectId('69f9b71982818aa2b309ad23')
mesin : "CNC-02"
suhu : 80.55
getaran : 0.12
timestamp : 2026-05-05T08:43:37.058+00:00
status : "normal"

_id: ObjectId('69f9b71982818aa2b309ad24')
mesin : "CNC-03"
suhu : 93
getaran : 0.48
timestamp : 2026-05-05T08:38:37.058+00:00
status : "normal"
```

4. Buat file read.py. Read: Query mesin CNC-01, tampilkan hanya suhu dan timestamp, urutkan terbaru, batasi 5.

```
from dotenv import load_dotenv
from pymongo import MongoClient
from datetime import datetime, timedelta
import random, os

load_dotenv()
client = MongoClient(os.getenv("MONGO_URI"))
db = client[os.getenv("DB_NAME")]
collection = db["sensor"]

cursor = collection.find(
    {"mesin": "CNC-01"},
    {"_id": 0, "suhu": 1, "timestamp": 1}
).sort("timestamp", -1).limit(5)
print("Data CNC-01 terbaru:")
for doc in cursor:
    print(doc)
```

Output:

```
PS C:\Users\mdhaf\Downloads\praktikum6> python -u "c:\Users\mdhaf\Downloads\praktikum6\read.py"
Data CNC-01 terbaru:
{'suhu': 64.0, 'timestamp': datetime.datetime(2026, 5, 5, 9, 23, 37, 58000)}
{'suhu': 97.24, 'timestamp': datetime.datetime(2026, 5, 5, 9, 13, 37, 58000)}
{'suhu': 69.95, 'timestamp': datetime.datetime(2026, 5, 5, 9, 3, 37, 58000)}
{'suhu': 90.12, 'timestamp': datetime.datetime(2026, 5, 5, 8, 58, 37, 58000)}
```

5. Update: Ubah status menjadi "maintenance" untuk semua mesin yang suhunya di atas 90.

```
from dotenv import load_dotenv
from pymongo import MongoClient
from datetime import datetime, timedelta
import random, os

load_dotenv()
client = MongoClient(os.getenv("MONGO_URI"))
db = client[os.getenv("DB_NAME")]
collection = db["sensor"]

update_result = collection.update_many(
    {"suhu": {"$gt": 90}},
    {"$set": {"status": "maintenance"}}
)
print(f"Update: {update_result.modified_count} dokumen diubah")
```

Output:

```
PS C:\Users\mdhaf\Downloads
Update: 1 dokumen diubah
```

6. Buat file delete.py. Delete: Hapus semua data yang lebih tua dari 1 jam yang lalu.

```
from dotenv import load_dotenv
from pymongo import MongoClient
from datetime import datetime, timedelta
import random, os

load_dotenv()
client = MongoClient(os.getenv("MONGO_URI"))
db = client[os.getenv("DB_NAME")]
collection = db["sensor"]

satu_jam_lalu = datetime.utcnow() - timedelta(hours=1)
delete_result = collection.delete_many({"timestamp": {"$lt":
satu_jam_lalu}})
print(f"Delete: {delete_result.deleted_count} dokumen dihapus")
```

Output:

```
(venv) PS C:\Users\mdhaf\Downloads\praktikum6> & c:/Users
c:\Users\mdhaf\Downloads\praktikum6\delete.py:11: Depreca
jects to represent datetimes in UTC: datetime.datetime.no
  satu_jam_lalu = datetime.utcnow() - timedelta(hours=1)
Delete: 0 dokumen dihapus
```

7. Buaat file agregasi.py lalu isi dengan kode berikut

```
import os
from dotenv import load_dotenv
from pymongo import MongoClient

load_dotenv()
client = MongoClient(os.getenv("MONGO_URI"))
db = client[os.getenv("DB_NAME")]
collection = db["sensor"]

pipeline = [
    {"$group": {
        "_id": "$mesin",
        "rata_suhu": {"$avg": "$suhu"},
        "max_getaran": {"$max": "$getaran"},
        "count": {"$sum": 1}
    }},
```

```
    {"$sort": {"rata_suhu": -1}}
]

results = collection.aggregate(pipeline)
print("Rata-rata suhu per mesin:")
for doc in results:
    print(f"{doc['_id']} -> rata:{doc['rata_suhu']:.2f}°C, max
getaran:{doc['max_getaran']}, jumlah data:{doc['count']}")

client.close()
```

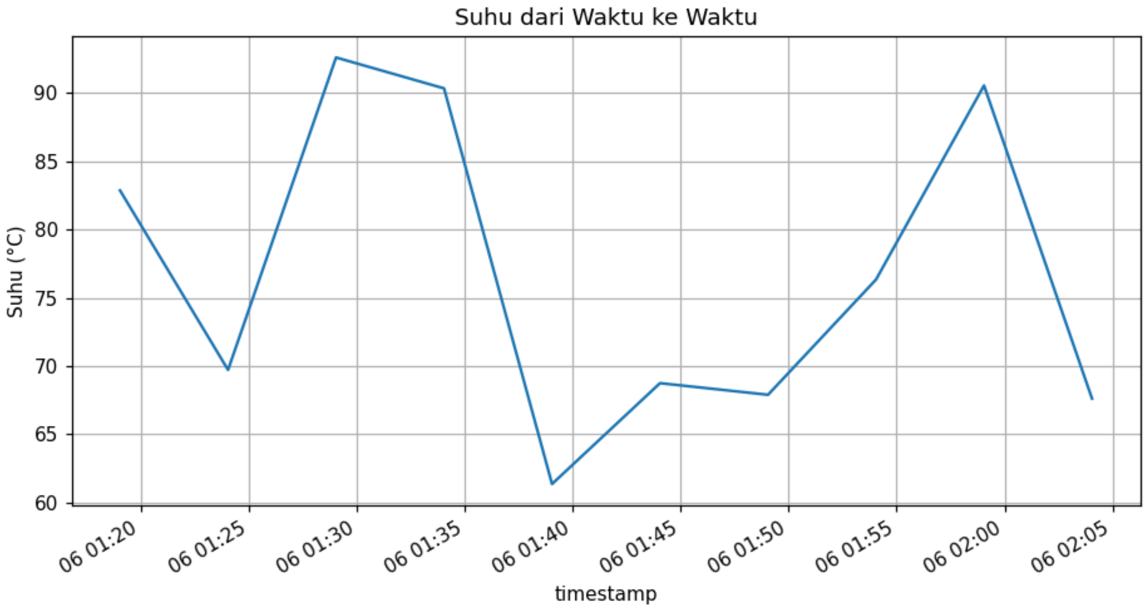
Output:

```
Rata-rata suhu per mesin:
CNC-01 -> rata:94.84°C, max getaran:0.45, jumlah data:1
CNC-02 -> rata:75.76°C, max getaran:0.31, jumlah data:4
CNC-03 -> rata:74.22°C, max getaran:0.46, jumlah data:5
```

8. Buat file analisis.py lalu isi dengan kode berikut

```
import os
import pandas as pd
import matplotlib.pyplot as plt
from dotenv import load_dotenv
from pymongo import MongoClient

load_dotenv()
client = MongoClient(os.getenv("MONGO_URI"))
db = client[os.getenv("DB_NAME")]
collection = db["sensor"]

cursor = collection.find({}, {"_id": 0})
df = pd.DataFrame(list(cursor))

df['timestamp'] = pd.to_datetime(df['timestamp'])
df.set_index('timestamp', inplace=True)

resampled = df.resample('10min').mean(numeric_only=True)
print(resampled.head())

plt.figure(figsize=(10,5))
df['suhu'].plot(title='Suhu dari Waktu ke Waktu')
plt.ylabel('Suhu (°C)')
plt.grid(True)
plt.savefig('suhu_plot.png')
plt.show()

client.close()
9.
```

Output:



10. Buat file data_generator.py dan diisi kode berikut

```
import time, random
from datetime import datetime
from pymongo import MongoClient
import os
from dotenv import load_dotenv

load_dotenv()
client = MongoClient(os.getenv("MONGO_URI"))
db = client[os.getenv("DB_NAME")]
collection = db["sensor"]

while True:
    doc = {
        "mesin": f"CNC-{random.randint(1,5):02d}",
        "suhu": round(random.uniform(60, 100), 2),
        "getaran": round(random.uniform(0.1, 0.5), 2),
        "timestamp": datetime.utcnow()
    }
    collection.insert_one(doc)
    print(f"[{datetime.utcnow()}] Inserted: {doc['mesin']} -
{doc['suhu']}°C")
    time.sleep(2)
```

```
[2026-05-05 13:36:11.097987] Inserted: CNC-01 - 63.58°C
[2026-05-05 13:36:13.099866] Inserted: CNC-02 - 74.4°C
[2026-05-05 13:36:15.101988] Inserted: CNC-04 - 96.93°C
[2026-05-05 13:36:17.104335] Inserted: CNC-02 - 71.3°C
[2026-05-05 13:36:19.107115] Inserted: CNC-05 - 74.57°C
[2026-05-05 13:36:21.108562] Inserted: CNC-04 - 88.92°C
[2026-05-05 13:36:23.112134] Inserted: CNC-02 - 74.12°C
[2026-05-05 13:36:25.115416] Inserted: CNC-01 - 94.77°C
[2026-05-05 13:36:27.119545] Inserted: CNC-02 - 65.63°C
[2026-05-05 13:36:29.123091] Inserted: CNC-05 - 93.57°C
[2026-05-05 13:36:31.125686] Inserted: CNC-02 - 75.15°C
[2026-05-05 13:36:33.128422] Inserted: CNC-05 - 75.76°C
[2026-05-05 13:36:35.131593] Inserted: CNC-05 - 70.83°C
```

11. Fase 5B (Lanjutan): Simulasi Data Sensor Melalui MQTT. Terminal 1: Publisher (mqtt_publisher.py)

```python
import paho.mqtt.client as mqtt
import json
from datetime import datetime
import time
import random

client = mqtt.Client(protocol=mqtt.MQTTv5)
client.connect("broker.hivemq.com", 1883, 60)

mesin_list = ["CNC-01", "CNC-02", "CNC-03", "CNC-04"]

while True:
    data = {
        "mesin": random.choice(mesin_list),             # random mesin
        "suhu": round(random.uniform(60, 100), 2),       # suhu random
        "getaran": round(random.uniform(0.1, 0.5), 2),   # getaran random
        "timestamp": datetime.now().isoformat()
    }

    client.publish("pabrik/sensor/suhu", json.dumps(data))
    print("🚀 Data dikirim:", data)

    time.sleep(2)
```

Output:

```
🚀 Data dikirim: {'mesin': 'CNC-01', 'suhu': 90.07, 'getaran': 0.33, 'timestamp': '2026-05-06T08:09:35.528341'}
🚀 Data dikirim: {'mesin': 'CNC-04', 'suhu': 97.23, 'getaran': 0.22, 'timestamp': '2026-05-06T08:09:37.531742'}
🚀 Data dikirim: {'mesin': 'CNC-02', 'suhu': 69.49, 'getaran': 0.44, 'timestamp': '2026-05-06T08:09:39.532599'}
🚀 Data dikirim: {'mesin': 'CNC-04', 'suhu': 61.48, 'getaran': 0.12, 'timestamp': '2026-05-06T08:09:41.533796'}
🚀 Data dikirim: {'mesin': 'CNC-02', 'suhu': 96.07, 'getaran': 0.25, 'timestamp': '2026-05-06T08:09:43.534886'}
🚀 Data dikirim: {'mesin': 'CNC-04', 'suhu': 99.28, 'getaran': 0.29, 'timestamp': '2026-05-06T08:09:45.536107'}
🚀 Data dikirim: {'mesin': 'CNC-03', 'suhu': 61.11, 'getaran': 0.15, 'timestamp': '2026-05-06T08:09:47.537044'}
```

12. Terminal 2: Subscriber (mqtt_subscriber.py)

```python
import json, os
import paho.mqtt.client as mqtt
from pymongo import MongoClient
from datetime import datetime
from dotenv import load_dotenv

load_dotenv()
MONGO_URI = os.getenv("MONGO_URI")
DB_NAME = os.getenv("DB_NAME")

mongo_client = MongoClient(MONGO_URI)
db = mongo_client[DB_NAME]
collection = db["sensor"]

def on_connect(client, userdata, flags, rc):
    print("Terhubung ke MQTT broker")
    client.subscribe("pabrik/sensor/suhu")

def on_message(client, userdata, msg):
    try:
        payload = json.loads(msg.payload.decode())
        payload['timestamp'] = datetime.fromisoformat(payload['timestamp'])
        result = collection.insert_one(payload)
        print(f"[MONGO] Tersimpan {result.inserted_id} - {payload['mesin']}
{payload['suhu']}°C")
    except Exception as e:
        print("Error:", e)

mqtt_client = mqtt.Client()
mqtt_client.on_connect = on_connect
mqtt_client.on_message = on_message
mqtt_client.connect("test.mosquitto.org", 1883, 60)
mqtt_client.loop_forever()
```

Output:

```
Tersimpan 69fa0334faf3289d83d23ad1 - CNC-05 90.06 C
Tersimpan 69fa0336faf3289d83d23ad2 - CNC-03 80.66 C
Tersimpan 69fa0338faf3289d83d23ad3 - CNC-02 63.56 C
Tersimpan 69fa033afaf3289d83d23ad4 - CNC-05 70.17 C
Tersimpan 69fa033cfaf3289d83d23ad5 - CNC-04 74.47 C
```

13. Fase 6: Error Handling dan Logging (10 menit)

Contoh penambahan logging pada crud_sensor.py (di awal file):

```
from dotenv import load_dotenv
from pymongo import MongoClient
from datetime import datetime, timedelta
import random, os

load_dotenv()
client = MongoClient(os.getenv("MONGO_URI"))
db = client[os.getenv("DB_NAME")]
collection = db["sensor"]
data = []
for i in range(10):
    doc = {
        "mesin": f"CNC-{random.randint(1,3):02d}",
        "suhu": round(random.uniform(60, 100), 2),
        "getaran": round(random.uniform(0.1, 0.5), 2),
        "timestamp": datetime.utcnow() - timedelta(minutes=i*5),
        "status": "normal"
    }
    data.append(doc)
result = collection.insert_many(data)
print(f"Tersimpan {len(result.inserted_ids)} dokumen")
```

Output:

```
c:\Users\mdhaf\Downloads\praktikum6\crud_sensor.py:16: Depreca
present datetimes in UTC: datetime.datetime.now(datetime.UTC).
  "timestamp": datetime.utcnow() - timedelta(minutes=i*5),
Tersimpan 10 dokumen
```

Kemudian bungkus insert_many dengan try-except:

```
from dotenv import load_dotenv
from pymongo import MongoClient
from datetime import datetime, timedelta
import random, os
import logging

load_dotenv()

client = MongoClient(os.getenv("MONGO_URI"))
db = client[os.getenv("DB_NAME")]
collection = db["sensor"]
```

```
# logging
logging.basicConfig(
    level=logging.INFO,
    format='%(asctime)s - %(levelname)s - %(message)s',
    handlers=[
        logging.FileHandler("crud_sensor.log"),
        logging.StreamHandler()
    ]
)


data = []
mesin_list = ["CNC-01", "CNC-02", "CNC-03"]

for i in range(10):
    data.append({
        "mesin": random.choice(mesin_list),
        "suhu": round(random.uniform(60, 100), 2),
        "timestamp": datetime.now() - timedelta(minutes=i)
    })

# insert ke MongoDB
try:
    result = collection.insert_many(data)
    logging.info(f"Insert berhasil, {len(result.inserted_ids)} dokumen")
except Exception as e:
    logging.error(f"Insert gagal: {e}")
```

Output:

```
(venv) PS C:\Users\mdhaf\Downloads\praktikum6> & c:/Users/mdha
2026-05-06 08:56:39,544 - INFO - Insert berhasil, 10 dokumen
```

Latihan Mandiri 1

1. Buat file csv lalu isi dengan ini:

```
mesin,tanggal,biaya,teknisi
CNC-01,2024-01-10,1200000,Andi
CNC-02,2024-01-15,800000,Budi
CNC-03,2024-01-20,600000,Joko
CNC-01,2024-02-05,1500000,Siti
CNC-02,2024-02-10,900000,Rina
CNC-03,2024-02-20,500000,Dedi
CNC-01,2024-03-12,2000000,Andi
CNC-02,2024-03-18,1100000,Budi
```

Setelah itu, save terlebih dahulu

2. Buat file import_maintenance.py dan isi dengan kode berikut:

```python
import pandas as pd
from pymongo import MongoClient

# Koneksi ke MongoDB
client = MongoClient("mongodb://localhost:27017/")
db = client["latihan6"]
collection = db["maintenance"]

# Baca file CSV
df = pd.read_csv("maintenance.csv", encoding="utf-8")

# Konversi kolom tanggal ke datetime
df["tanggal"] = pd.to_datetime(df["tanggal"])

# Ubah ke list of dictionary
data = df.to_dict(orient="records")

# Insert ke MongoDB
if data:
    collection.insert_many(data)
    print("✅ Data berhasil diimport!")
else:
    print("❌ Data kosong, tidak ada yang diinsert")
```

Output:

```
PS C:\Users\mdhaf\Downloads\l
n6/import_maintenance.py
✅ Data berhasil diimport!
```

3. Buat file query_maintenance.py lalu isi dengan kode berikut:

```
import pandas as pd
from pymongo import MongoClient

# Koneksi ke MongoDB
client = MongoClient("mongodb://localhost:27017/")
db = client["latihan6"]
collection = db["maintenance"]

# =========================
# 1. Query biaya > 1.000.000
# =========================
result = list(collection.find(
    {"biaya": {"$gt": 1000000}},
    {"_id": 0}
))

df = pd.DataFrame(result)

print("=== Data biaya > 1.000.000 ===")
print(df)


# =========================
# 2. Update data
# =========================
update_result = collection.update_one(
    {"mesin": "CNC-01", "biaya": 1200000},
    {"$set": {"teknisi": "Dewi"}}
)

print("\n=== Update ===")
print("Matched:", update_result.matched_count)
print("Modified:", update_result.modified_count)


# =========================
# 3. Aggregation total biaya per bulan
# =========================
pipeline = [
    {
        "$project": {
            "bulan": {
                "$dateToString": {
                    "format": "%Y-%m",
                    "date": "$tanggal"
                }
            },
```

```python
                "biaya": 1
            }
        },
        {
            "$group": {
                "_id": "$bulan",
                "total_biaya": {"$sum": "$biaya"}
            }
        },
        {
            "$sort": {"_id": 1}
        }
]

agg_result = list(collection.aggregate(pipeline))

df_agg = pd.DataFrame(agg_result)

if not df_agg.empty:
    df_agg.rename(columns={"_id": "bulan"}, inplace=True)

print("\n=== Total biaya per bulan ===")
print(df_agg)
```

Output:

```
PS C:\Users\mdhaf\Downloads\latihan6> & C:
n6/query_maintenance.py
=== Data biaya > 1.000.000 ===
    mesin    tanggal    biaya teknisi
0  CNC-01 2024-01-10  1200000    Andi
1  CNC-01 2024-02-05  1500000    Siti
2  CNC-01 2024-03-12  2000000    Andi
3  CNC-02 2024-03-18  1100000    Budi

=== Update ===
Matched: 1
Modified: 1

=== Total biaya per bulan ===
     bulan  total_biaya
0  2024-01      2600000
1  2024-02      2900000
2  2024-03      3100000
PS C:\Users\mdhaf\Downloads\latihan6>
```

Latihan Mandiri 2 (Opsional)

1. Buat file mqtt_publisher.py

```
import paho.mqtt.client as mqtt
import json
import time
import random

broker = "broker.hivemq.com"
topic = "pabrik/produksi"

client = mqtt.Client()
client.connect(broker, 1883, 60)

mesin_list = ["CNC-01", "CNC-02", "CNC-03", "CNC-04"]
batch_list = ["B001", "B002", "B003", "B004"]

while True:
    data = {
        "batch": random.choice(batch_list),
        "mesin": random.choice(mesin_list),
        "jumlah": random.randint(100, 500),
        "reject": random.randint(0, 50)
    }

    client.publish(topic, json.dumps(data))
    print("Kirim:", data)

    time.sleep(3)
```

Output:

```
PS C:\Users\mdhaf\Downloads\latihan6> & C:/Users/mdhaf/AppData/Local/Pytho
n6/mqtt_publisher.py
c:\Users\mdhaf\Downloads\latihan6\mqtt_publisher.py:9: DeprecationWarning:
  client = mqtt.Client()
Kirim: {'batch': 'B001', 'mesin': 'CNC-03', 'jumlah': 252, 'reject': 50}
Kirim: {'batch': 'B003', 'mesin': 'CNC-03', 'jumlah': 386, 'reject': 39}
Kirim: {'batch': 'B004', 'mesin': 'CNC-04', 'jumlah': 132, 'reject': 47}
Kirim: {'batch': 'B001', 'mesin': 'CNC-01', 'jumlah': 403, 'reject': 35}
Kirim: {'batch': 'B001', 'mesin': 'CNC-04', 'jumlah': 490, 'reject': 47}
```

2. Buat file mqtt_subscriber.py

```
import paho.mqtt.client as mqtt
import json
from pymongo import MongoClient
from datetime import datetime

mongo_client = MongoClient("mongodb://localhost:27017")
db = mongo_client["latihan_mqtt"]
collection = db["produksi_mqtt"]

broker = "broker.hivemq.com"
topic = "pabrik/produksi"

def on_connect(client, userdata, flags, rc):
    print("Terhubung ke broker")
    client.subscribe(topic)

def on_message(client, userdata, msg):
    data = json.loads(msg.payload.decode())

    data["timestamp"] = datetime.now()

    reject_rate = (data["reject"] / data["jumlah"]) * 100
    data["reject_rate"] = reject_rate

    if reject_rate > 5:
        print("⚠ PERINGATAN! Reject rate terdeteksi tinggi:",
reject_rate)
        data["peringatan"] = True
    else:
        data["peringatan"] = False

    collection.insert_one(data)

    print("Data masuk:", data)

client = mqtt.Client()
client.on_connect = on_connect
client.on_message = on_message

client.connect(broker, 1883, 60)
client.loop_forever()
```

Output:

```
PS C:\Users\mdhaf\Downloads\latihan6> & C:/Users/mdhaf/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/mdhaf/Downloads/latiha
n6/mqtt_subscriber.py
c:\Users\mdhaf\Downloads\latihan6\mqtt_subscriber.py:35: DeprecationWarning: Callback API version 1 is deprecated, update to latest versi
on
  client = mqtt.Client()
Terhubung ke broker
⚠PERINGATAN! Reject rate terdeteksi tinggi: 16.39344262295082
Data masuk: {'batch': 'B002', 'mesin': 'CNC-03', 'jumlah': 183, 'reject': 30, 'timestamp': datetime.datetime(2026, 5, 6, 10, 18, 17, 9258
06), 'reject_rate': 16.39344262295082, 'peringatan': True, '_id': ObjectId('69fab2f9ecb8fa8bd44547e9')}
Data masuk: {'batch': 'B004', 'mesin': 'CNC-02', 'jumlah': 223, 'reject': 5, 'timestamp': datetime.datetime(2026, 5, 6, 10, 18, 20, 92141
4), 'reject_rate': 2.242152466367713, 'peringatan': False, '_id': ObjectId('69fab2fcecb8fa8bd44547ea')}
⚠PERINGATAN! Reject rate terdeteksi tinggi: 25.396825396825395
Data masuk: {'batch': 'B004', 'mesin': 'CNC-02', 'jumlah': 189, 'reject': 48, 'timestamp': datetime.datetime(2026, 5, 6, 10, 18, 23, 9253
69), 'reject_rate': 25.396825396825395, 'peringatan': True, '_id': ObjectId('69fab2ffecb8fa8bd44547eb')}
Data masuk: {'batch': 'B003', 'mesin': 'CNC-02', 'jumlah': 465, 'reject': 3, 'timestamp': datetime.datetime(2026, 5, 6, 10, 18, 26, 92621
4), 'reject_rate': 0.6451612903225806, 'peringatan': False, '_id': ObjectId('69fab302ecb8fa8bd44547ec')}
```